

Applied Analysis 7

Rohan Hore

August 2024

1 Antibody activity of lupus patients in response to the influenza virus

Lupus is an autoimmune disease, which can affect any part of the body. As with other autoimmune diseases such as diabetes, the immune system attacks the body's own cells, causing inflammation and tissue damage. The full name *Systemic lupus erythematosus*, is abbreviated to **SLE**.

This question is based on data from ten lupus and 8 control patients who were given an influenza vaccine in tightly monitored circumstances. Lupus and control patients were selected at random from a pool of patients in a larger study in the U.S., and asked to take part in this influenza study. The measurements were made on those who gave informed consent; no information is available on the participation rate. The body responds to the vaccine by producing antibodies, a very large number of particles varying from one patient to another. The study aims to determine if the antibodies produced by SLE patients differ from those produced by the controls, and if so, in what ways. In immunology, an 'individual antibody' is not a single particle, but a monoclonal batch of particles identical at the molecular level. Each measurement on an individual antibody is a bulk measurement made on a monoclonal batch generated by a single cell extracted from the serum of one individual. Each antibody has an affinity value for the influenza virus, which varies from antibody to antibody and patient to patient.

The file `lupus.dat` contains information for 255 antibodies taken from the 18 patients. Of the 11 variables, the first five `sex`, `age`, `race`, `sle`, and `pid` (patient identity) are self-explanatory. The primary response `vaffinity` (antibody-virus affinity) is a measure of the antibody reaction to the influenza virus: this number is ordinarily in the range $10^6 - 10^{10}$ in standard conventional units. For computational convenience the value given here has been re-scaled by multiplying by 10^{10} . The influenza virus comes in several strains, and each antibody ordinarily reacts to a specific strain; `vaffinity` is the maximum of the strain-specific affinities. The strain to which each antibody reacts is recorded in `vstrain`, a 7-level factor, and the virus type in `vtype`, whose levels are **B**, **H1** and **H3**.

The number of antibodies per patient ranges from five to 43. One possibility is that these numbers are random as a consequence of laboratory protocols related to aliquot volumes and dilution levels selected by the experimenter independently of serum content. Another possibility is that antibody production levels vary randomly from patient to patient, and that these sample antibody numbers are roughly proportional to the individual profusions. Although the former scenario seems more plausible, some of the questions asked below are phrased as if the latter holds. In answering such questions, you should explain the implications of one scenario versus the other, and tailor your answers accordingly.

A certain fraction of the antibodies also react to another protein, called **RHA**. The active antibodies and their **RHA** affinity values are given in the variable `rhaff`, also multiplied by 10^{10} .

Two further quantitative variables, the affinity values for two proteins **HA1** and **MN**, are also reported. Although formal statistical models may be useful to address some of the following questions, you should bear in mind that simple graphs and tables of averages may be equally compelling, especially if accompanied by a relevant standard error.

Data Reading

```
lupus_data=read.table("lupus.dat.txt",header=TRUE)
head(lupus_data)
```

##	sex	age	race	sle	pid	vaffinity	rhaff	HA1	MN	vtype	vstrain
## 1	F	31	C	sle	SLE1	3100.00	0.791	NA	NA	B	MAL
## 2	F	31	C	sle	SLE1	45.70	3.487	NA	NA	H1	SI
## 3	F	31	C	sle	SLE1	209.00	28.390	NA	NA	H1	SI
## 4	F	31	C	sle	SLE1	4130.00	2582.000	NA	NA	H3	WIS
## 5	F	31	C	sle	SLE1	9.72	3.174	1.875	0.3703704	B	MAL
## 6	F	43	C	sle	SLE2	1.57	NA	NA	NA	H3	WIS

2 Possible Questions

Problem 1

In terms of the descriptors commonly applied to scientific studies, comparative, randomized, replicated, case-control, and so on, how would you describe this study? Justify your answer. **True/False:** In terms of partitions `vtype` produces larger partitions compared to `vstrain`?

Answer: (observational study, true)

```
attach(lupus_data)
sle=as.factor(sle);sex=as.factor(sex);race=as.factor(race)
```

Problem 2

Are the control patients appropriately selected? Do the controls differ systematically from the SLE patients in ways that are unrelated to the disease but could affect antibody activity comparisons?

Answer:

```
par(mfrow=c(1,3))
boxplot(age~sle)
hist(age[sle=="sle"],xlab="age",main="age distribution of sle group")
hist(age[sle=="cntrl"],xlab="age",main="age distribution of control group")

#female proportions
par(mfrow=c(1,1))
length(which(sex=="F" & sle=="sle"))/length(which(sle=="sle"))
length(which(sex=="F" & sle=="cntrl"))/length(which(sle=="cntrl"))

#C race proportions
length(which(race=="C" & sle=="sle"))/length(which(sle=="sle"))
length(which(race=="C" & sle=="cntrl"))/length(which(sle=="cntrl"))
```

female proportions in sle group and control group: 0.641,0.597.
C race proportions in sle group and control group: 0.763,0.653.

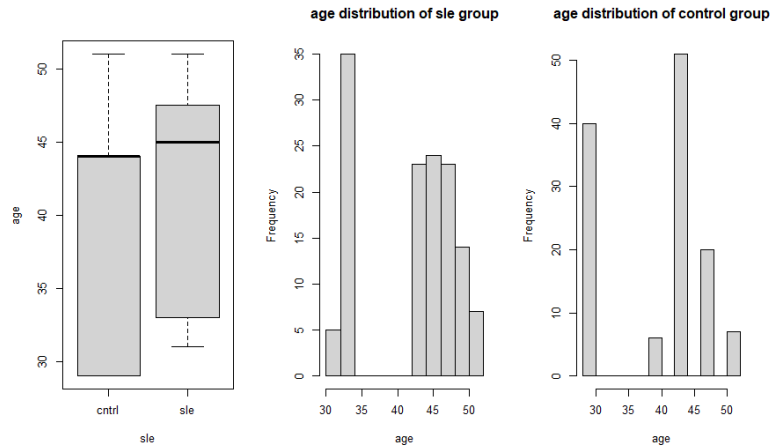


Figure 1: Plot 1

Problem 3

One group of key scientific questions is concerned with how and whether the antibodies produced by SLE patients are different from those produced by the controls, either in number of antibodies produced or in viral activity levels. **vaffinity** is the main variable that we are going to study for that analysis. What kind of data-preprocessing do you think is necessary before doing any inference?

Answer:

```
hist(vaffinity)
```

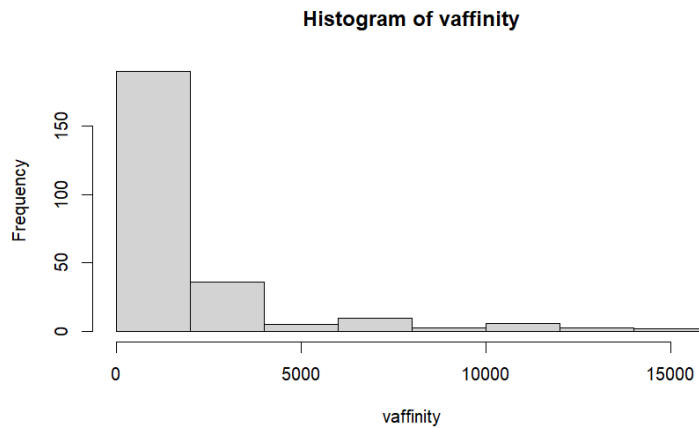


Figure 2: Plot 2

```
#a simple solution: log transformation
par(mfrow=c(1,2))
hist(vaffinity);hist(log(vaffinity))
```

Alternative solution: box-cox transformation

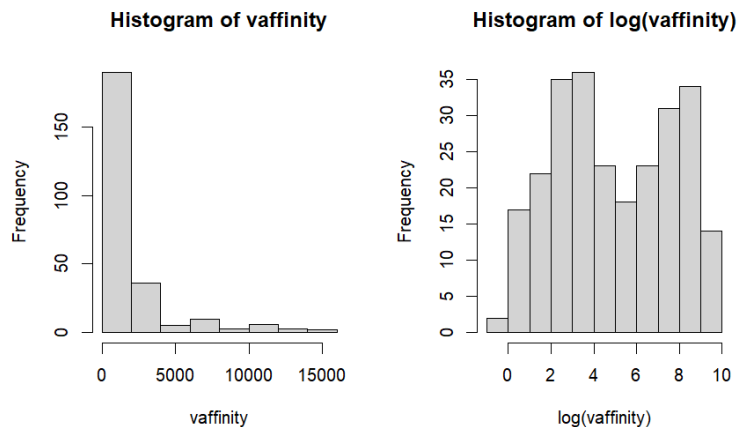


Figure 3: Plot 3

Problem 4

Do the SLE antibodies have greater or lesser viral affinity than the controls? Explain carefully what accommodations, if any, are made for the other variables sex, age etc.

Answer:

```
#ignoring presence of features
t.test(log(vaffinity)[sle=="sle"],log(vaffinity)[sle=="cntrl"],paired=FALSE,
alternative="two.sided")

#a better choice: an intuitive model with random effects for each patient
library(nlme)
model1=lme(log(vaffinity)~sex+age+race+sle,data=lupus_data,random=~1|pid)
summary(model1)
```

One solution could be ignoring all features and do a T-test to study the following hypothesis test

$$H_0 : \mu_{\text{vaff,control}} = \mu_{\text{vaff,SLE}} \quad \text{vs} \quad H_1 : \text{not } H_0$$

```
## Welch Two Sample t-test
##
## data: log(vaffinity)[sle == "sle"] and log(vaffinity)[sle == "cntrl"]
## t = -2.9347, df = 252.99, p-value = 0.003646
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
## -1.6381014 -0.3224373
## sample estimates:
## mean of x mean of y
## 4.480958 5.461227
```

A more thoughtful answer would be fitting a linear mixed model on log-transformed vaffinity with sex, age, race, grouping and we add a random effect of patient id. For patient i , observation j

$$\log(\text{vaffinity}_{ij}) = \alpha \text{sex}_i + \beta \text{age}_i + \gamma \text{race}_i + \delta 1_{\{i \in \text{SLE}\}} + \text{patient}_i + \epsilon_{ij}; \quad \text{var}(\epsilon_{ij}) = \sigma_\epsilon^2, \text{var}(\text{patient}_i) = \sigma_p^2$$

```
## Linear mixed-effects model fit by REML
## Data: lupus_data
##      AIC      BIC    logLik
## 1208.445 1233.095 -597.2226
```

```
##
## Random effects:
## Formula: ~1 | pid
## (Intercept) Residual
## StdDev: 0.9057422 2.44263
##
## Fixed effects: log(vaffinity) ~ sex + age + race + sle
## Value Std.Error DF t-value p-value
## (Intercept) 4.197505 1.8621494 237 2.2541185 0.0251
## sexM -1.746747 0.7416112 13 -2.3553406 0.0349
## age 0.056599 0.0424928 13 1.3319561 0.2058
## raceC -0.228115 0.8012069 13 -0.2847143 0.7803
## slesle -1.201690 0.5725851 13 -2.0987103 0.0559
## Correlation:
## (Intr) sexM age raceC
## sexM 0.088
## age -0.884 -0.385
## raceC -0.493 0.409 0.104
## slesle 0.071 0.055 -0.258 -0.010
##
## Standardized Within-Group Residuals:
## Min Q1 Med Q3 Max
## -2.15848559 -0.65116926 -0.02248569 0.68136442 2.17876421
##
## Number of Observations: 255
## Number of Groups: 18
```

The fitted responses for SLE on average are 1.2 smaller than the control group. The evidence of SLE effect is borderline; not too strong but suggestive. The effect of age and race is not significant while male patients have significantly a lower mean as compared to female patients.

Problem 5

Is viral affinity related to the number of antibodies sampled? Compute a relevant test statistic with explanation as needed. How do you interpret this effect, if any?

Answer:

```
#number of antibodies for each patient
nantibodies=rep(0,18)
k=1
for(id in unique(pid)){
  nantibodies[k]=length(which(pid==id))
  k=k+1
}
#a new data frame
lupus_data1=cbind(lupus_data,0)
colnames(lupus_data1)=c(colnames(lupus_data), "nantibodies")
k=1
for(id in unique(pid)){
  lupus_data1$nantibodies[lupus_data1$pid==id]=nantibodies[k]
  k=k+1
}

#effect of nantibodies
model2=lme(log(vaffinity)~sex+sle+nantibodies,data=lupus_data1,random=~1|pid)
summary(model2)
```

Since, age and race were insignificant in the previous model and only retain sex and the grouping effect while

also adding the number of antibodies in the model.

$$\log(\text{vaffinity}_{ij}) = \alpha \text{sex}_i + \delta 1_{\{i \in \text{SLE}\}} + \eta_i \text{ #antibodies}_i + \text{patient}_i + \epsilon_{ij}; \quad \text{var}(\epsilon_{ij}) = \sigma_\epsilon^2, \text{var}(\text{patient}_i) = \sigma_p^2$$

```
## Linear mixed-effects model fit by REML
## Data: lupus_data1
##      AIC      BIC    logLik
## 1209.173 1230.326 -598.5866
##
## Random effects:
## Formula: ~1 | pid
##      (Intercept) Residual
## StdDev:   0.8664064 2.443891
##
## Fixed effects:  log(vaffinity) ~ sex + sle + nantibodies
##               Value Std.Error DF   t-value p-value
## (Intercept)  6.602894 0.6072816 237 10.872871  0.0000
## sexM         -0.946141 0.6241752  14 -1.515825  0.1518
## slesle        -1.129009 0.5472950  14 -2.062889  0.0582
## nantibodies  -0.033701 0.0274660  14 -1.227023  0.2400
## Correlation:
##      (Intr) sexM   slesle
## sexM      0.019
## slesle    -0.586 -0.129
## nantibodies -0.710 -0.370  0.184
##
## Standardized Within-Group Residuals:
##      Min      Q1      Med      Q3      Max
## -2.21268508 -0.63349491 -0.02520506  0.67885332  2.18902548
##
## Number of Observations: 255
## Number of Groups: 18
```

The effect of nantibodies is estimated as -0.033701 (standard error: 0.027). The p-value suggests the effect is not significant. There are two scenarios: one is laboratory level randomness. Under this scenario, even if the effect was significant it would be hard to interpret. While, in the second scenario, where it's individual level randomness, a significant effect is much easily interpretable.

Problem 6

Do SLE patients produce antibodies in greater or lesser number than the controls? Explain your calculations and your reasoning.

Answer:

```
mean(nantibodies[1:10]) - mean(nantibodies[11:18])
t.test(nantibodies[1:10], nantibodies[11:18], paired=FALSE, alternative = "two.sided")
```

Mean difference between two groups: -2.4. We are assuming two things here: (1) the number of antibodies recorded is a good indication (proportional to) number of antibodies produced. (2) there is no effect of demographics as such on the number of antibodies produced.

With these, we can simply implement a t-test for the equality of group means

```
##
## Welch Two Sample t-test
##
```

```
## data:  nantibodies[1:10] and nantibodies[11:18]
## t = -0.46958, df = 11.785, p-value = 0.6472
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -13.558437    8.758437
## sample estimates:
## mean of x mean of y
##      13.1      15.5
```

Note:- Small sample size: an issue?

Problem 7

Not all antibodies react to the RHA protein. Let p_0 be the fraction of control antibodies that exhibit RHA activity, and let p_1 be the corresponding parameter for SLE patients. By using a Bayesian or other argument, compute the probability

$$\mathbb{P}(p_1 \geq p_0 | \text{data})$$

How will numerically compute it?

Answer: Let p_i be the fraction of antibodies that exhibit RHA activity for i th individual. Let's say $p_i \stackrel{d}{=} p_j$ if i, j have same SLE status. Suppose, NA_i is the number of antibodies recorded for individual i , while NRH_i is the number of antibodies that exhibit RHA activity. Clearly, for all i , $NRH_i \sim \text{Bin}(NA_i, p_i)$. We can put a prior on p_i , treating it as random. Let's say

$$p_i \sim \begin{cases} \text{Beta}(\alpha_0, \beta_0) & \text{if } i \text{ is in control group} \\ \text{Beta}(\alpha_1, \beta_1) & \text{if } i \text{ is in SLE group} \end{cases}$$

Hence, marginally

$$NRH_i \sim \begin{cases} \text{beta-binomial}(NA_i, \alpha_0, \beta_0) & \text{if } i \text{ is in control group} \\ \text{beta-binomial}(NA_i, \alpha_1, \beta_1) & \text{if } i \text{ is in SLE group} \end{cases}$$

Is $\alpha_0 = \alpha_1$ and $\beta_0 = \beta_1$?

```
n_RHA=rep(0,18)
k=1
for(id in unique(pid)){
  n_RHA[k]=length(which(pid==id & is.na(rhaff)==FALSE))
  k=k+1
}

library(VGAM)
group=c(rep("sle",10),rep("cntrl",8))
fit_combined=vglm(cbind(n_RHA, nantibodies-n_RHA) ~1, betabinomialff, trace = TRUE)
exp(coef(fit_combined,matrix=TRUE))

fit_group=vglm(cbind(n_RHA, nantibodies-n_RHA) ~group, betabinomialff, trace = TRUE)
est=exp(coef(fit_group,matrix=TRUE))
rbind(est[1,],est[1,]*est[2,])

#LRT
2*(logLik(fit_group)-logLik(fit_combined))
```

LRT test statistic: 0.9213

Based on above LRT test, since the group effect is not too significant, we can treat $\pi_0 \stackrel{d}{=} \pi_1$ and then the estimate of the probability will be simply 0.5. Otherwise, we can do a numerical simulation to find the probability when the (α, β) are different for both groups.

```
alpha_0=5.057;beta_0=2.588
alpha_1=26.705;beta_1=11.382
pi_0=rbeta(5000,shape1=alpha_0,shape2=beta_0)
pi_1=rbeta(5000,shape1=alpha_1,shape2=beta_1)
sum(pi_1>=pi_0)/5000
```

The simulation result estimates the probability to be 0.5472.

Problem 8

The individual antibody affinity measurements are expensive and time-consuming. Nevertheless, the researcher is considering an extension of the experiment in order to pin down more accurately the difference in activity levels between SLE and control patients. The following proposals are similar in terms of cost:

1. Make 5 additional antibody measurements on each of the current patients, thereby enlarging the total number of measurements from 255 to 345.
2. Select four additional patients, two controls and two SLE, and make five antibody affinity measurements on each.

Assuming that all parameter values remain unchanged, which of the two scenarios lead to more reduction in the variance of the SLE effect?